

Security Doorbell Cam — Cheat Sheet

Chain K — Hardware & Systems Foundations · one-page recall, not first-time learning

Camera & Motion Detection

Task	Rule
Fix PIR false triggers	Debounce (min HIGH duration) + cooldown after each event
PIR buying red flag	No adjustable sensitivity/delay pots

Edge Pipeline & Notifications

Fact	Detail
Pipeline shape	Sensor trigger → local inference/decision → notification
Latency target	Under ~2 seconds, end to end
Blown budget, fast network/capture	Fix the inference/decision stage — it's the bottleneck

Two-Way Audio

Fact	Detail
Feedback cause	Speaker output re-entering the mic and re-amplifying
Feedback fix	Physical separation between speaker and mic

Storage & Weatherproofing

Fact	Detail
Storage fix	Rolling policy — max size/days, delete oldest first
Recording mode	Motion-triggered over continuous — less storage, less SD wear
Camera weatherproofing	Sealed clear (never tinted) acrylic window
PIR weatherproofing	Must stay exposed (gasket-ringed) — needs direct IR line-of-sight